

Manu Shrivastava

|332 Adarsh Nagar, Post Basharatpur Gorakhpur U.P 273004 | shrivastava.manu87@gmail.com | +91-8290670273 |
www.shrivastavamanu.co.in

Summary

I am a Computer Science academic and researcher with over 9 years of teaching experience and a PhD from Kyushu Institute of Technology, Japan. A recipient of Japan's prestigious MEXT (Japanese Government) Scholarship. My research expertise lies in Natural Language Processing (NLP), Quantum Computing with a particular focus on Neural Machine Translation, ontology population and optimization. My doctoral research explored the development of improved Neural Machine Translation approaches for ontology population tasks, including the application of innovative loss functions and statistical analysis techniques.

My academic interests and expertise encompass Natural Language Processing, Machine Learning, Artificial Intelligence, Neural Machine Translation, and related areas of Computer Science. In addition to my academic qualifications, I am an Oracle Certified Java Programmer. I have been actively involved in teaching, research, student mentoring, and academic activities, with a commitment to contributing to the academic and research objectives of the institution.

EXPERIENCE

Assistant Professor

Jul. 2013 – Oct. 2021

Manipal University Jaipur - self-financed state private university.

Jaipur, India

- Responsible for teaching subjects such as Object-Oriented Programming using JAVA, DBMS (Database Management Systems), and DAA (Design and Analysis of Algorithms)

Assistant Controller of Examination (Additional Responsibility)

Manipal University Jaipur - self-financed state private university.

Jaipur, India

- Contributed to the administration by overseeing examination processes, ensuring smooth coordination and timely execution of exams.

Assistant Professor

Jan. 2025 – Present

Manipal University Jaipur - self-financed state private university.

Jaipur, India

- PhD Coordinator
- Member Board of Studies
- Member Department Research Committee
- High Performance Computing Lab Faculty Coordinator

EDUCATION

Kyushu Institute of Technology

Kitakyushu, Japan

Doctor of Philosophy in Engineering, Department of Life Science and systems engineering

2021 - 2024

Title: The Effect of the Grammatical Complexity of Natural Language Descriptions in the Translation to Semantic Triples by Sequential Models: Its Performance and Internal Mechanisms.

Supervisor: Prof. Hiroaki Wagatsuma, Professor, KYUTECH, Japan.

Description- My work addresses the challenge of improving Neural Machine Translation (NMT) for ontology population tasks, i.e., extracting concepts and relationships from natural language text. The research identifies the impact of sentence structure on the performance of sequential models, a factor often overlooked in existing approaches. The thesis proposed a **Modified Jeffreys Divergence (MJD)** loss function, which combines the mean-seeking behavior of forward Kullback-Leibler Divergence (KLD) and the mode-seeking behavior of backward KLD, leading to improved accuracy in concept and relation extraction.

The thesis provides statistical insights using methods such as the **Kolmogorov-Smirnov test** and the **Cramer Von Mises test** to assess the learning behavior of models trained on different sentence structures. The findings emphasize the importance of sentence structures in training data to enhance NMT performance in real-world applications. The thesis introduced a **Structure Dependent Weighted Loss Function** that adjusts learning

weights based on sentence types, ensuring better generalization across different sentence structures. The work also includes a **Conditional Checkpoint Selection Strategy**, which dynamically selects model checkpoints based on sentence type during training, ensuring optimal performance across diverse sentence structures. Ultimately, my contributions lead to more robust and accurate knowledge extraction for ontology population tasks.

Malaviya National Institute of Technology

Master of Technology, Department of computer engineering

Jaipur, India

2011 - 2013

Title: Ahead of Time Compiler.

Supervisor: Dr. Dinesh Gopalani Associate Professor MNIT Jaipur, India.

Description: This work focuses on optimizing Android applications by utilizing native code for performance-critical tasks. We developed standard methods in both Java and native languages (C/C++) and compared their performance, including parallel versions using OpenMP, on a Dual-core ARM Cortex-A9 processor. To further enhance efficiency, we developed a prototype converter that transforms DEX code into C code for performance-critical “hot” methods, which are then compiled into native code using GCC. The native code is integrated with the application using the Java Native Interface (JNI), demonstrating a pathway for optimizing Android applications for better computational efficiency..

Institute of Engineering and Technology

Bachelor of Technology, Department of Computer Science and Engineering

Indore, India

2005 - 2009

Project Title: Online Port Scanner.

Description: The project aims to provide an online tool for networking such as a port scanner, whois query, etc. The main aim of the project is to provide an alternative to users for networking tools as many of the corporate systems block such commands on their system.

Technical Skills

Programming Languages: C, C++, JAVA, PYTHON, Android App Development.

Certification:MOOC

1. Bayesian Statistic: From Concept to Data Analysis by University of California Santa Cruz.
2. Convolution Neural Network, DeepLearning.AI.
3. Improving Deep Neural Network: Hyperparameter tuning, Regularization and optimization. DeepLearning.AI.
4. Neural Network and Deep Learning, DeepLearning.AI.
5. Structuring Machine learning Projects DeepLearning.AI.
6. Machine learning foundations: A case study approach, University of Washington.
7. Oracle-certified Java programmer.
8. Introduction to Retrieval Augmented Generation (RAG)

Workshop/Conference Participation

1. Participant in 20 days FDP on QT-05 quantum Computation by EICT Academy MNIT, December 08-31, 2025.
2. Participant in 2 days Workshop on Pedagogic Effectiveness by MUJ TEC, July 22-23, 2016.
3. Member Organizing Committee in Fifth International Conference on Innovations in Computational Intelligence and Computer Vision (ICICV 2025) June 04-06, 2025
4. Participant in The 17th International Conference on Innovative Computing, Information and Control (ICI-CIC2023), Kumamoto, Japan, August 29-31, 2023.
5. Participant in 14th International Conference, ICSR 2022, Florence, Italy, December 13–16, 2022.
6. Member Organizing Committee 1st International Conference on Smart Systems, Innovations and Computing, Manipal University Jaipur 2017.
7. Participant in 5 days FDP on Computational Intelligence By Manipal University Jaipur Jan 2017.
8. Participant in 10 days GIAN on Public Key Infrastructure and Trust Management by NIT Goa Nov 2016.
9. Participant in 3 days Workshop on High Power Computing By CDAC Jan 2016.
10. Participant in 5 days FDP on Wireless Networks by the Department of Computer Science and Engineering MNIT Jaipur, Jan 2016.
11. Participant in 3 days FDP by Manipal Center for Professional and Personal Development. May 2014.
12. Participant in Two days Professor Summit on Embedded System Design using Atmel XMEGA A3BU and ARM-based D20, Dec 2014.

Publications

- J1 Shrivastava, Manu, Kosei Shibata, and Hiroaki Wagatsuma. Structure dependent weighted loss function for triple extraction using BART model. Pattern Analysis and Applications 29, no. 1 (2026): 37.
- J2 Shrivastava M. Shibata, K., & Wagatsuma, H.: Conditional checkpoint selection strategy based on sentence structures for text to triple translation using BiLSTM encoder–decoder model. Int J Data Sci Anal (2024).
- J3 Shrivastava M. and Wagatsuma H.: A Quality Analysis of the BiLSTM Encoder Decoder Model with Modified J-Divergence for Sentences with Different Complexities. ICIC Express Letters, Part B: Applications (ICIC-ELB) 2023, Volume 18, Number 7, pp 721-729.
- J4 Dixit, A., Oshiumi, W., Shrivastava, M., & Wagatsuma, H. (2024). An image processing mechanism for aerial inspection robots to detect submillimeter-width concrete cracks in social infrastructures. Advanced Robotics, 38(9–10), 698–714.
- J5 Banerjee, S., Singal, G., Saha, S., Mittal, H., Srivastava, M., Mukherjee, A., ... & Garg, D. (2022). Machine Learning approach to Predict net radiation over crop surfaces from global solar radiation and canopy temperature data. International Journal of Biometeorology, 66(12), 2405-2415.
- J6 Shrivastava, M., Roy, S., Kumar, K., Pandey, C. V., & Grover, J. (2021). LICCA: a lightweight image cipher using 3-D cellular automata. Nonlinear Dynamics, 106, 2679-2702.
- J7 Roy, S., Shrivastava, M., Rawat, U., Pandey, C. V., & Nayak, S. K. (2021). IESCA: An efficient image encryption scheme using 2-D cellular automata. Journal of Information Security and Applications, 61, 102919.
- J8 Roy, S., Shrivastava, M., Pandey, C. V., Nayak, S. K., & Rawat, U. (2021). IEVCA: An efficient image encryption technique for IoT applications using 2-D Von-Neumann cellular automata. Multimedia Tools and Applications, 80, 31529-31567.

- C1 Shrivastava, M., Seri, K., & Wagatsuma, H. (2022, December). A Named Entity Recognition Model for Manufacturing Process Based on the BERT Language Model Scheme. In International conference on social robotics (pp. 576-587). Cham: Springer Nature Switzerland.

Patent

- P1 "A METHOD AND SYSTEM FOR AI-BASED ADAPTIVE TUTORING THROUGH ARCHITECTURE DESIGN"- Application No. 202511043833 INDIA Date of filing of Application: 06/05/2025 Publication Date: 23/05/2025.
- P2 "SYSTEM AND METHOD FOR SOFTWARE IMPLEMENTATION OF AN AUTONOMOUS AI TUTOR WITH REAL-TIME OCR AND FEEDBACK." Application No. 202511043832, INDIA Date of filing of Application: 06/05/2025 Publication Date: 23/05/2025.

References

Prof. Hiroaki Wagatsuma

Professor, Department of Human Intelligence Systems, Graduate School of Life Science and Systems Engineering, Kyushu Institute of Technology (KYUTECH), Japan.

Dr. Dinesh Gopalani

Associate Professor, Department of Computer Science and Engineering, Malaviya National Institute of Technology, Jaipur.